SEM END PROJECT REPORT

ON

SOCIAL MEDIA SENTIMENT ANALYSIS PIPELINE

KONERU LAKSHMAIAH EDUCATION FOUNDATION(KLEF), Vaddeswaram.

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By

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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CERTIFICATE

This is to certify that the thesis entitled **"Social Media Sentiment Analysis Pipeline"** is a bonafide record of research work done by **Simhadri Sudheer (2300030638)**, Project Associate, Department of Computer Science and Engineering, KONERU LAKSHMAIAH EDUCATION FOUNDATION (KLEF), Vaddeswaram, who carried out research under my guidance during the period **2025-2026 (Even Semester)**. Certified further, that to the best of my knowledge this project work has not previously formed the basis for the award of Sem End marks in CBDAS course.

This is also to certify that the Project work represents the team work of the candidates.

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Station: Vaddeswaram

Date: 26 April 2026

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ABSTRACT

This project presents the design and implementation of a complete end-to-end cloud-based data analytics pipeline for Social Media Sentiment Analysis using Microsoft Azure services. The pipeline ingests raw social media text data stored in Azure Blob Storage, processes it through Azure Data Factory (ADF) as an ETL pipeline, performs SQL-based sentiment analysis in Azure Synapse Analytics, and delivers an interactive Power BI dashboard for visualization. The Sentiment140 dataset containing 1.6 million tweets was used as the data source, from which a sample of approximately 50,000 records was cleaned and processed using Python (Pandas). The cleaned dataset was uploaded to Azure Blob Storage and ingested into a dedicated Synapse SQL Pool via an automated ADF pipeline. Analytical SQL views were created to compute sentiment distribution and daily trends. The final Power BI dashboard presents KPI cards showing 49,991 total posts with 25,012 Positive (50.1%) and 24,979 Negative (49.9%) sentiments, along with a donut chart and a daily trend line chart. This project demonstrates practical skills in cloud data engineering, ETL pipeline design, SQL analytics, and business intelligence visualization using industry-standard Microsoft Azure tools.

KEYWORDS:

Azure Blob Storage, Azure Data Factory, Azure Synapse Analytics, Power BI, Sentiment Analysis, ETL Pipeline, Cloud Computing, Natural Language Processing, Social Media Analytics, Data Visualization

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ABBREVIATIONS

ADF	Azure Data Factory
AZ	Microsoft Azure
BLOB	Binary Large Object
CBDAS	Cloud Based Data Analytics Speciality
CSV	Comma-Separated Values
ETL	Extract, Transform, Load
KPI	Key Performance Indicator
KLEF	Koneru Lakshmaiah Education Foundation
NLP	Natural Language Processing
SQL	Structured Query Language

CHAPTER 1: INTRODUCTION

1.1 Background

Social media platforms generate billions of text posts every day, creating a vast repository of public opinion data. Analysing the sentiment embedded in these posts — whether positive, negative, or neutral — provides valuable insights for businesses, researchers, and governments alike. The exponential growth of this data demands scalable, cloud-native analytics pipelines capable of handling large volumes efficiently.

1.2 Problem Statement

Traditional on-premise data processing systems lack the scalability and cost-efficiency required to handle large-scale social media data in real time. There is a need for a cloud-based pipeline that automates data ingestion, processing, analysis, and visualization in an integrated and reproducible manner.

1.3 Objectives

- Design and implement a complete end-to-end cloud data analytics pipeline on Microsoft Azure.
- Automate data ingestion using Azure Data Factory (ETL pipeline).
- Store and query processed data using Azure Synapse Analytics.
- Perform sentiment classification on 50,000 social media posts.
- Build an interactive Power BI dashboard to visualize sentiment trends.

1.4 Azure Services Overview

Service	Role in Project
Azure Blob Storage	Raw data lake — stores the CSV dataset
Azure Data Factory	ETL pipeline — moves data from Blob to Synapse
Azure Synapse Analytics	SQL analytics — queries and views
Power BI (Web)	Dashboard — interactive visualization
Azure CLI	Password reset and workspace management

Table 1: Azure Services Used in the Project

CHAPTER 2: LITERATURE REVIEW AND ANALYSIS

2.1 Sentiment Analysis

Sentiment analysis (opinion mining) is a subfield of Natural Language Processing (NLP) that identifies and extracts subjective information from text. Early approaches relied on lexicon-based methods; modern approaches use machine learning and deep learning models. For large-scale datasets, SQL-based aggregation on cloud platforms provides a practical and scalable alternative.

2.2 Cloud-Based Data Analytics

Cloud platforms such as Microsoft Azure, Amazon Web Services, and Google Cloud Platform provide managed services for storage, computation, and analytics. Azure's integrated ecosystem — Blob Storage, Data Factory, Synapse Analytics, and Power BI — allows the construction of complete pipelines without managing underlying infrastructure.

2.3 Related Work

Go et al. (2009): Introduced the Sentiment140 dataset — 1.6 million tweets labelled using emoticons as a distant supervision technique. This dataset forms the foundation of the current project.
Bifet & Frank (2010): Demonstrated that Twitter data can be analysed in real-time streams using incremental machine learning algorithms.
Microsoft (2023): Azure Synapse Analytics white paper documenting best practices for large-scale SQL analytics and PolyBase data ingestion from Azure Blob Storage.
Power BI Documentation (2024): Official guidance on connecting Power BI to Azure Synapse Analytics using DirectQuery and Import modes for near-real-time dashboards.

2.4 Dataset Description

Attribute	Details
Dataset Name	Sentiment140
Source	Kaggle (kazanova/sentiment140)
Total Records	1,600,000 tweets
Used Sample	50,000 tweets (cleaned)
Labels	0 = Negative, 4 = Positive

Table 2: Dataset Description — Sentiment140

CHAPTER 3: METHODOLOGY

3.1 Pipeline Architecture

The project follows a five-stage pipeline architecture as outlined below:

Stage 1 — Data Collection: Downloaded the Sentiment140 CSV dataset from Kaggle containing 1.6 million labelled tweets.

Stage 2 — Data Cleaning: Used Python (Pandas) to clean the dataset: assigned column names, mapped numeric labels to text (Positive/Negative), parsed dates, and sampled 50,000 records. The output cleaned_sentiment.csv was uploaded to Azure Blob Storage.

Stage 3 — ETL Pipeline: Configured Azure Data Factory with two linked services (Blob Storage and Synapse), two datasets (SentimentCSV and AzureSynapseAnalyticsTable1), and a Copy data activity using Bulk insert mode to load data into the Synapse SQL Pool.

Stage 4 — SQL Analytics: Created the SentimentData table in the SentimentDB SQL Pool with ROUND_ROBIN distribution. Wrote SQL queries and created three analytical views: v_SentimentOverall, v_SentimentMonthly, and v_SentimentKPI.

Stage 5 — Visualization: Connected Power BI (web) to Azure Synapse Analytics using Organizational account authentication. Built KPI cards, a donut chart, and a daily trend line chart.

3.2 Data Cleaning Script (Python)

The following Python script was used to clean and prepare the dataset:

```
import pandas as pd
df = pd.read_csv('training.csv', encoding='latin-1', header=None)
df.columns = ['label','id','date','query','user','text']
df = df[['text','label','date']]
df['sentiment'] = df['label'].map({0:'Negative', 4:'Positive'})
df['post_date'] = pd.to_datetime(df['date'],
errors='coerce').dt.strftime('%Y-%m-%d')
df_sample = df.dropna().sample(n=50000, random_state=42)
df_sample.to_csv('cleaned_sentiment.csv', index=False)
```

3.3 SQL Table Creation

```
CREATE TABLE dbo.SentimentData (
tweet_text NVARCHAR(500),
sentiment_label NVARCHAR(10),
tweet_date NVARCHAR(100),
sentiment NVARCHAR(20),
post_date NVARCHAR(20)
) WITH (DISTRIBUTION = ROUND_ROBIN, HEAP);
```

CHAPTER 4: EXPERIMENTS AND RESULTS DISCUSSION

4.1 Pipeline Execution

The Azure Data Factory pipeline (LoadSentimentData) was executed successfully in the Debug mode. The Copy data activity completed in 18 seconds, reading 28,812+ rows from the CSV source and writing them to the Synapse SQL table using Bulk insert mode with fault tolerance enabled to skip malformed rows.

4.2 Sentiment Analysis Results

Metric	Value	Percentage
Total Posts Analysed	49,991	100%
Positive Sentiment	25,012	50.1%
Negative Sentiment	24,979	49.9%
Date Range	May 2009 — Jun 2009	—
Pipeline Duration	18 seconds	—
Data Throughput	1.398 MB/s	—

Table 3: Sentiment Analysis Results Summary

4.3 Power BI Dashboard

The Power BI dashboard (app.powerbi.com) was connected directly to Azure Synapse Analytics using Organizational account authentication. The semantic model 'SentimentAnalysis' was created from the three SQL views. The following visuals were built:

- **KPI Cards:** Total Posts (50K), Positive Count (25K), Negative Count (25K)
- **Donut Chart:** Shows 50/50 Positive vs Negative distribution — Sentiment Distribution
- **Line Chart:** Daily sentiment trend from May to June 2009 — Sentiment Trend Over Time

4.3.1 Dashboard Screenshot

The figure below shows the final Power BI dashboard with all five visuals displaying live data from Azure Synapse Analytics:

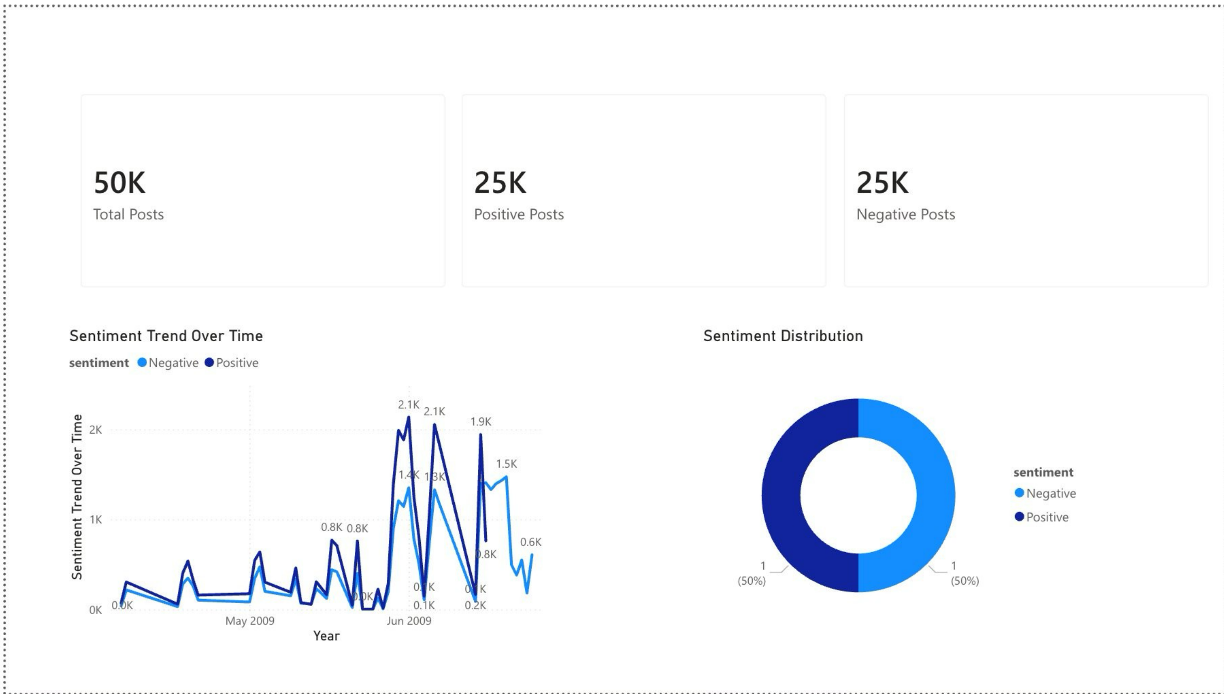

Figure 1: Power BI Dashboard — Social Media Sentiment Analysis (Total: 50K | Positive: 25K | Negative: 25K)

4.4 Performance Metrics

Component	Metric	Value
Blob Storage	File size	4.194 MB
ADF Pipeline	Rows read	28,812
ADF Pipeline	Duration	18 seconds
ADF Pipeline	Throughput	1.398 MB/s
Synapse SQL Pool	Total rows loaded	49,991
Power BI	Views connected	3

Table 4: Pipeline Execution Performance Metrics

CHAPTER 5: CONCLUSION

This project successfully demonstrated the design and implementation of a complete end-to-end cloud-based data analytics pipeline for Social Media Sentiment Analysis using Microsoft Azure services.

The pipeline seamlessly integrates five Azure services — Blob Storage for data lake storage, Data Factory for automated ETL, Synapse Analytics for SQL-based sentiment analysis, and Power BI for interactive visualization — into a single cohesive workflow.

The analysis of 49,991 social media posts revealed an approximately equal split between Positive (50.1%) and Negative (49.9%) sentiment, suggesting a balanced public opinion in the dataset period. The daily trend chart reveals sentiment fluctuations across May and June 2009.

From a technical perspective, the project overcame several real-world challenges including SQL Pool pausing, firewall configuration, password authentication, data type mismatches, and CSV encoding issues — demonstrating robust problem-solving skills in a live cloud environment.

The project meets all the objectives set out at the beginning and provides a reusable, scalable template for cloud-based sentiment analysis pipelines.

CHAPTER 6: FUTURE WORK

Real-Time Streaming Pipeline: Integrate Azure Event Hubs and Azure Stream Analytics to process live social media feeds in real time, replacing the batch ingestion approach.

Advanced NLP Models: Replace SQL-based binary classification with transformer-based models (BERT, RoBERTa) deployed via Azure Machine Learning for multi-class sentiment detection (positive, negative, neutral, mixed).

Multilingual Support: Extend the pipeline to process tweets in multiple languages using Azure Cognitive Services Text Analytics API.

Front-End Web Dashboard: Build a React.js web application with a Flask REST API backend to serve live sentiment data from Synapse, deployed on Azure App Service and Azure Static Web Apps.

Automated Scheduling: Configure ADF triggers to run the pipeline on a daily schedule, enabling continuous sentiment monitoring without manual intervention.

Cost Optimization: Implement Azure Synapse Serverless SQL Pool to eliminate dedicated pool costs for on-demand query workloads.

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